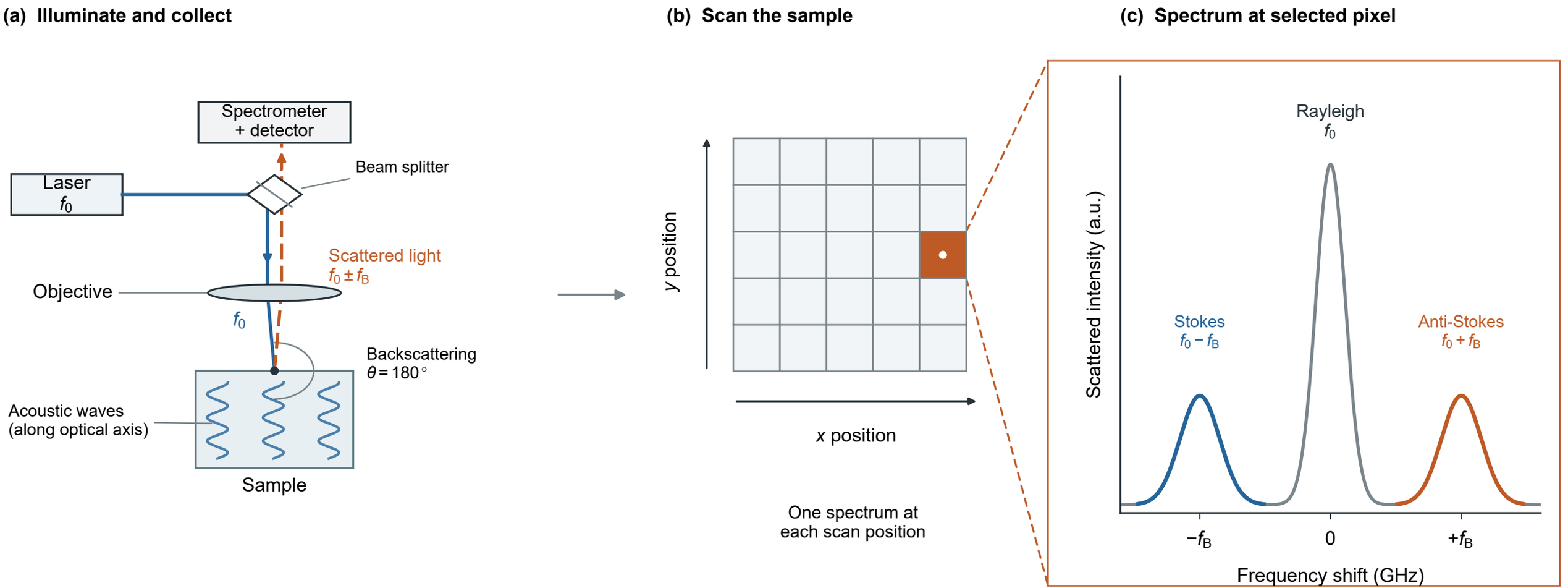

(a) Illuminate and collect
Spectrometer + detector
Beam splitter
Laser f_0
Scattered light $f_0 \pm f_B$
Objective
f_0
Backscattering $\theta = 180°$
Acoustic waves (along optical axis)
Sample
(b) Scan the sample
y position
x position
One spectrum at each scan position
(c) Spectrum at selected pixel
Rayleigh f_0
Stokes $f_0 - f_B$
Anti-Stokes $f_0 + f_B$
Scattered intensity (a.u.)
$-f_B$
0
$+f_B$
Frequency shift (GHz)